

Methodology Report:

Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	Duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	Id	Unique identifier
	Method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	Tripid	Unique identifier of trips
	loc_from	Source Location code
	Searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trips	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount
	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID
	custid	Customer ID
	distance	Distance in KM from source to destination
	duration	Unique identifier of time periods like duration_id

Points to Note:

1. Without this methodology document, the other parts of your case study will not be evaluated.
2. This assignment is different from the ones you have solved before. Make sure that you treat this case study as a storytelling exercise and not an analysis/visualisation one. This will help you be better prepared for the presentations.
3. Once you are done with the analysis and visualisations, there will be many insights at your hand. Make sure that you map the right visuals and takeaways with the right audience since some of these insights might be relevant to one group but not to the other group.
4. DO NOT change the text or numbering of any task, as it may cause problems with grading. Write your solutions to a task in the space provided below the respective task.

Tasks to be performed

- Present the overall approach of the analysis.
- Mention the problem statement and the analysis approach briefly.
- To solve a task, you have to create relevant visualisations and derive appropriate insights from the visualisations.
- Add all the plots, insights, calculated field commands, results and outcomes for a task with proper numbering and sequence in the report.
- The scores for all tasks (except conclusions) comprise both analysis work in the visualisation tool and its outcome in the report.
- You will be awarded a score for a task only if the Tableau/PowerBI analysis is correct and is included in the report along with the subsequent insights.
- Finally, draw conclusions based on the analysis.

Scoring:

Report Total Marks: 70

Sections: 3 sections (10 marks + 40 marks + 20 marks)

Analysis and Visualisation

1. Data Preparation

[10 Marks]

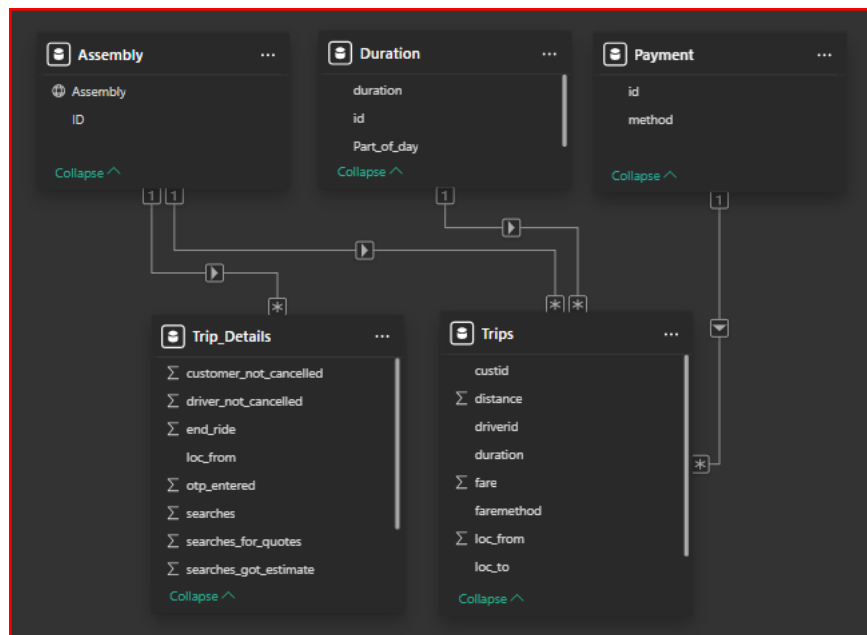
1.1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

Solution:

All tables were imported into **Power BI query editor**. Using Column quality and Column distribution options, analytically tried to explore errors, duplicates and data type of each column in all the tables. Ensured relevant headers are promoted in each table before loading to Model view.

We identified two tables' **Trips** and **Trips_Details** as fact tables and **Assembly**, **Duration** and **Payment** tables as dimension tables. We created fact constellation schema, where we both the facts tables are sharing the data from one or many dimension tables. Refer figure-1 for the schema: -



(Figure-1)

We established one-to-many cardinality between the tables to ensure data integrity, allowing for accurate data aggregation and meaningful analysis. **Refer figure- 2**

☐ From: table (column) ↑	Relationship	To: table (column)	Status
☐ Trip_Details (loc_from)	* — 1	Assembly (ID)	Active
☐ Trips (duration)	* — 1	Duration (id)	Active
☐ Trips (faremethod)	* — 1	Payment (id)	Active
☐ Trips (loc_to)	* — 1	Assembly (ID)	Active

(Figure-2)

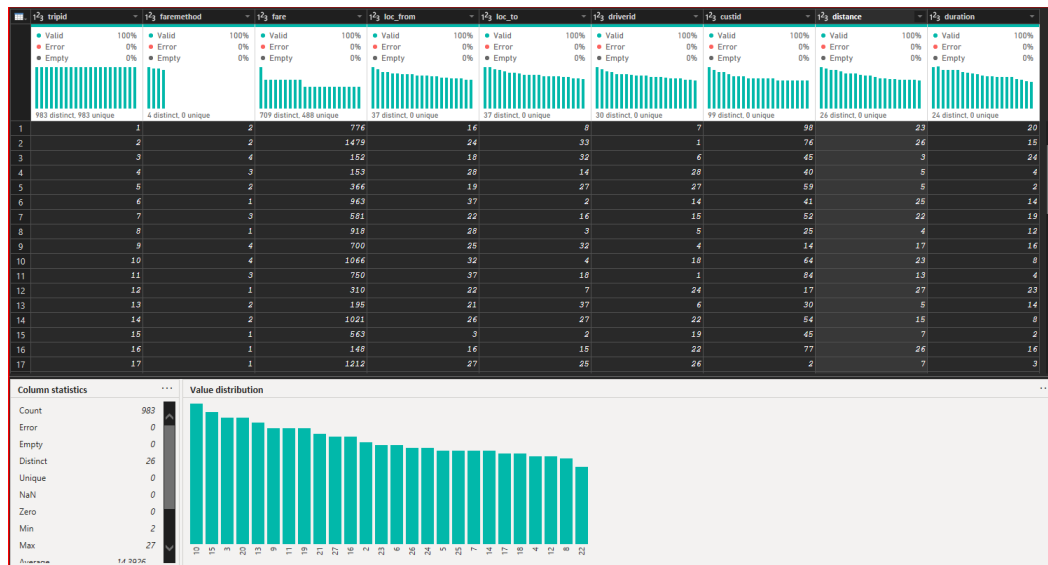
1.2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might affect the analysis.
- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution:

To find inconsistencies we followed various approaches: -

1. In Table view, we applied logical filters to counts and find inconsistent values, Missing values at Random and duplicate vales.
2. In Power Query window, we did column profiling based on entire data set. Figure-3 shows column statistics, which we evaluated



(Figure-3)

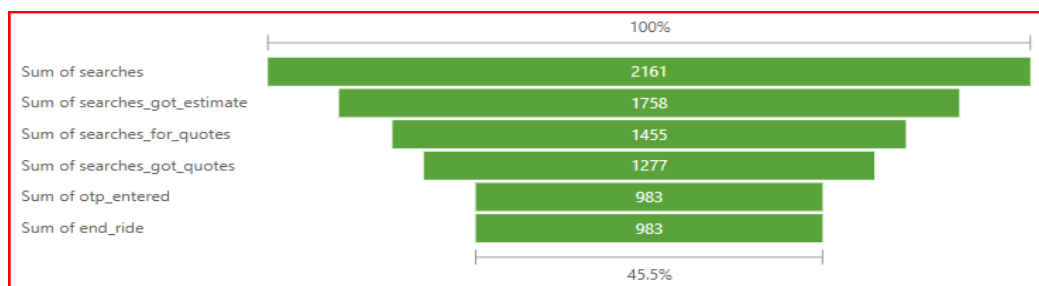
To resolve inconsistencies, we followed following approach: -

1. **Missing Values:** There were no missing values in the dataset.
2. **Duplicates:** The Primary keys in all the table were unique. No duplicate values found.
3. **Data Type Consistency:** In power query, we ensured correct data type is assigned to the corresponding data. Example -numerical fields such as searches, searches_got_estimate in Trip_Detail table are Whole number (Integer) and that categorical fields in Payment table such as method as Test data type (strings).
4. **Logical Consistency:** Checked and found that the number of values in the columns of Trip_Details followed logical constraints. e.g., $\text{end_rides} \leq \text{otp_entered} \leq \text{searches_got_quotes} \leq \text{searches_for_quotes} \leq \text{searches_got_estimate} \leq \text{Searches}$. We verified the distribution using a Funnel chart.

Refer

figure-4:

-



(Figure-4)

We created one **calculated column** in duration table- **Part_of_day** (Conditional column) to bin duration into Mid Night (00-03 Hrs.), Early Morning (03-07 Hrs.), Forenoon (07-12 Hrs.), Afternoon (12-17 Hrs.), Evening (17-20 Hrs.) and rest as Night. **The M-Code snippet**

```
= Table.AddColumn("#Changed Type",  
"Part_of_day",  
each if [id] <= 3 then "Mid Night"  
else if [id] >= 4 and [id] <8 then "Early Morning "  
else if [id] >= 8 and [id] <13 then "Forenoon"  
else if [id] >= 13 and [id] <18 then "Afternoon"  
else if [id] >= 18 and [id] <21 then "Evening"  
else if [id] >= 21 then "Night"  
else null)
```

2. Exploratory Data Analysis

[40 Marks]

2.1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution:

Table Name	Column Name	Data Type
Assembly	Assembly_ID	Categorical
	Assembly	Categorical
Duration	duration_id	Categorical
	Duration	Categorical
Payment	Id	Categorical
	Method	Categorical
Trip Details	Tripid	Categorical
	loc_from	Categorical
	Searches	Categorical
	searches_got_estimate	Categorical
	searches_for_quotes	Categorical
	searches_got_quotes	Categorical

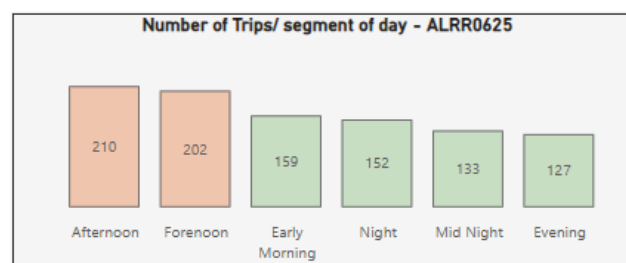
	customer_not_cancelled	Categorical
	driver_not_cancelled	Categorical
	otp_entered	Categorical
	end_ride	Categorical
Trips	tripid	Categorical
	faremethod	Categorical
	fare	Numerical
	loc_from	Categorical
	loc_to	Categorical
	driverid	Categorical
	custid	Categorical
	distance	Numerical
	duration	Categorical

2.2. Analyse Ride Demand Over Time [3 Marks]

- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

Solution:

To determine the ride distribution, we summarized number of rides completed over segment of the day (Calculated column) and then further drilled down to analyze hourly distribution along with distribution of average fare. Peak demand recorded during afternoon followed by Forenoon. Refer figure-5

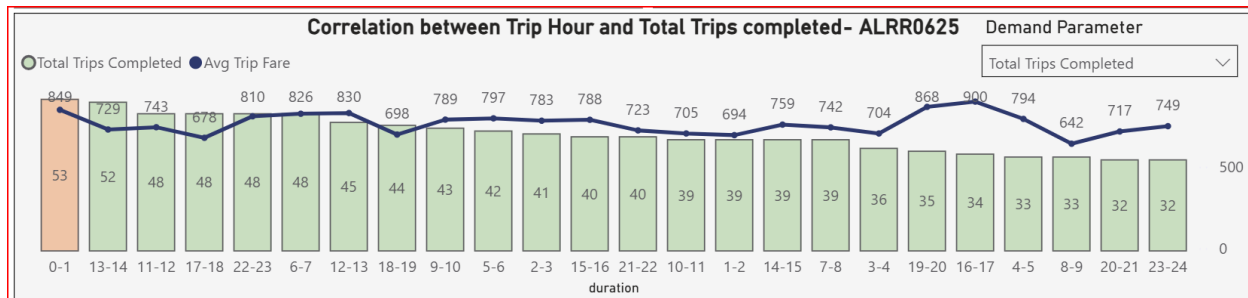


(Figure-5)

Following hours within each segment of the day (refer figure- 6) recorded

highest number of trips: -

- Midnight: 00-01 Hrs.
- Afternoon: 13-14 Hrs.
- Morning: 11-12 Hrs.
- Evening: 17-18 Hrs.
- Night: 22-23 Hrs.



(Figure-6)

We used the following DAX to derive Average trip fare: -

```
Avg Trip Fare = sum(Trips[fare]) / (COUNTROWS(Trips))
```

2.3. Proportion of Total Revenue from Different Time Periods [3 Marks]

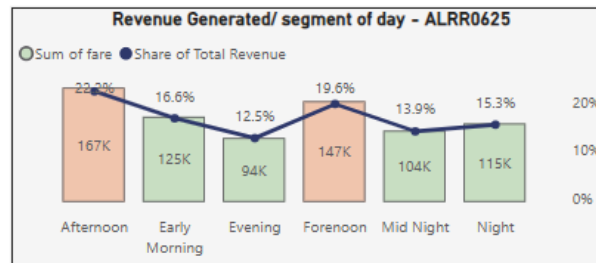
- Calculate the proportion of revenue generated during different time periods and visualise how it contributes to total revenue.

Solution:

To calculate proportion of revenue generated during different period, we used the following DAX: -

```
Share of Total Revenue = DIVIDE(
    SUM(
        Trips[fare]),
    CALCULATE(SUM(Trips[fare]), ALL(trips))
)
```


And then cumulative share of the revenue per segment of the day can be visualized using figure-7



(Figure-7)

2.4. Explore the Relationship Between Trip Hour and Revenue [3 Marks]

- Investigate the correlation between trip hour and total fare.
- Explain any trends or patterns that emerge.

Solution:

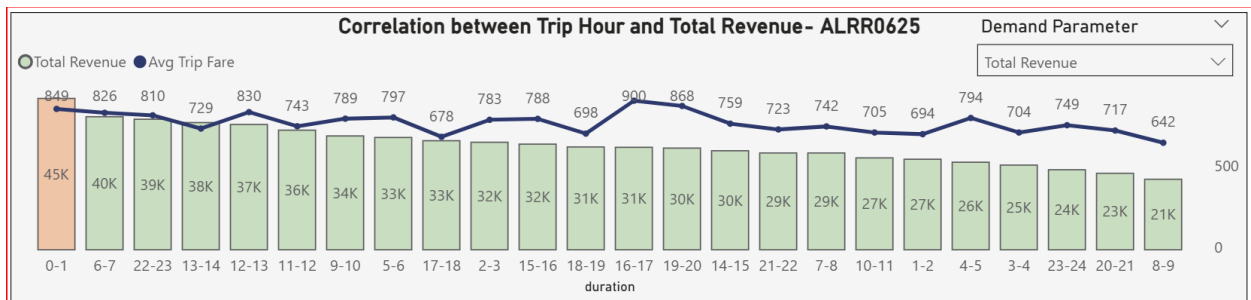
To find correlation between trip hour and revenue, we called out Total Revenue DAX code within *Demand Parameter*: -

```
Total Revenue = sum(Trips[fare])

Demand Parameter = {
    ("Total Trips Completed", NAMEOF('Measure_Table'[Total
Trips Completed]), 0),
    ("Total Revenue", NAMEOF('Measure_Table'[Total Revenue]),
1)
}
```

Top revenue generating Hours:

- 0–1 AM — ₹45,019 (~6.0%)
- 6–7 AM — ₹39,624 (~5.3%)
- 0–11 PM (22–23) — ₹38,880 (~5.2%)
- 1–2 PM (13–14) — ₹37,897 (~5.0%)
- 12–1 PM (12–13) — ₹37,341 (~5.0%)



(Figure-8)

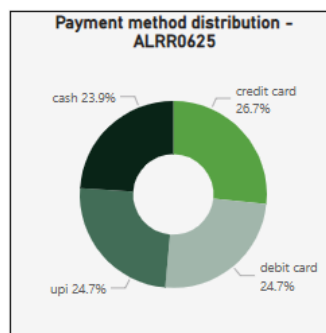
Late-night (0–1 AM) and early morning (6–7 AM) rides have high revenue contributing ~11% of the total, suggesting surge pricing or longer trips. Evening and lunchtime slots also contribute significantly, likely from urban office travel and food delivery routes.

2.5. Examine the Popularity of Different Payment Methods [3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

Solution:

We used **Donut Chart** to find the most preferred/ common payment method used by the customers. Refer figure-9



(Figure-9)

Digital payments dominate, with nearly 76% of rides using Credit/Debit Cards or UPI. Credit cards are slightly more popular, potentially due to reward points or corporate usage. Cash usage is lowest, but still significant—suggesting the need to retain support for offline payments, especially for older or non-digital users.

2.6. Identify High-Performing Zones [6 Marks]

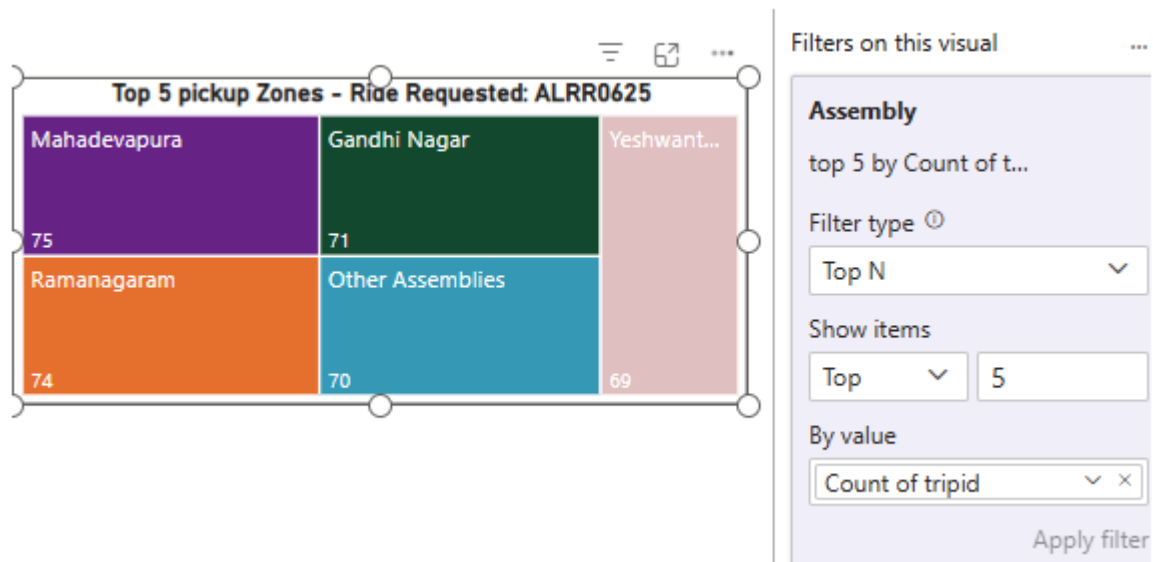
Identify zones with the highest number of rides and revenue generation.
Analyse factors contributing to their performance:

- 2.6.1. Rides: Identify pickup zones with the highest number of trip requests.
[3 marks]

Solution:

To identify pickup zones with the highest number of trips requested, we selected assemblies (Pick-up locations) from Assembly table and applied **Top N filter** on number of rides recorded in Trips_Details table.

We used Tree plot visual and applied visual level filter to find the following locations with count of rides requested (refer figure -10): -



(Figure-10)

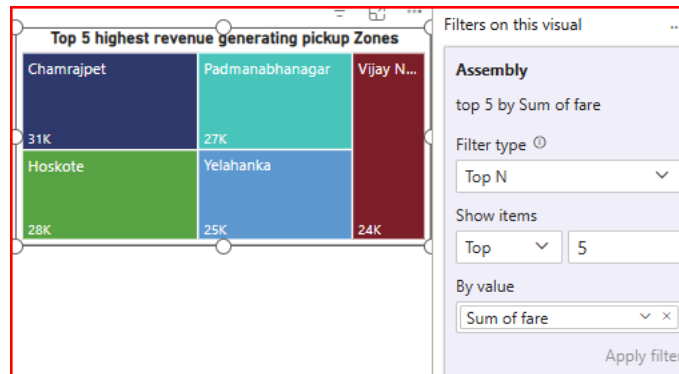
- 2.6.2. Revenue: Identify pickup zones generating the highest revenue.
[3 marks]

Solution:

To identify pickup zones generating the highest revenue, we selected assemblies (Pick-up locations) from Assembly table and applied **Top N filter** on number of fare (sum of fare) recorded in Trips table.

We used Tree plot visual and applied visual level filter to find the following

locations with highest number of revenue generated (refer figure -11): -



(Figure-11)

2.7. Analyse Ride Time Periods Across Zones [4 Marks]

- Compare the trip trends for different time periods across pickup zones.

Solution:

To compare trip trends for different time periods across all the assemblies (pick-up zones), we used a **Matrix** visual with Assemblies in Rows, duration (time period) in columns and number of trips (tripid) in values.

Further we applied **Conditional Formatting** on number of trips using *rule-based* approach to highlight background to indicate magnitude of rides completed.

Background color - Count of tripid

Format style: Rules Apply to: Values only

What field should we base this on? Count of tripid Summarization: Count

Rules

If value	then
is blank	
>= 1 Number and < 2 Number	
>= 2 Number and <= 3 Number	
>= 4 Number and < 10 Number	

Reverse color order New rule

(Figure-12)

Green shades represent higher numbers of rides and red shades represent zero or less numbers of rides. Amber is moderate number of rides.

Assembly	0-1	10-11	11-12	1-2	12-13	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22-23	2-3	23-24	3-4	4-5	5-6	6-7	7-8	8-9	9-10
Anekal	1	1	1	1		3				1	1	3	1	1	1	1	1	1	1	1	3	2	1	1
B. T. M. Layout	1	2		3	1	1		1		2	1	1		1	2	2	1	1	2	1		2	2	2
Bangalore South	1	1	1		2		1		4			1	1	1	2			1			2		2	1
Basavanagudi	1				1	1		1	1	1		2	1	1	1		1	2	1	2	1		1	
Bommanahalli		1	3		1	4	1	1	1	1		1		1	1				1	2	2	1		3
Byatarayanapura	1		2		1	2					1	2		2	2	1	1	1	1				2	
C. V. Raman Nagar			1		2		2	1	1	3	2				1	1	1	1	1	1		2		2
Chamrajpet	5	2	1	2		1	5	1	1	4		1	1	1	2			2		2	2	2		1
Channarayana	1	1	3		1	2	1		2			2			1	2	2	1				2	1	1
Chickpet		1	2	1	1		1		2			1		1	3		1	2	2	4	1			2
Dasarahalli	1		2	1	2	2	3		1	3	2			2	2		1	1		3	1	2	1	1
Devanahalli	2	1			1	2		1		1	1	3	2	2	2	1	2	2	2	2	1	1		1
Doddaballapur		1	1	2			1	2	1	2	1				1	1	2	3		3	2	2	2	
Gandhi Nagar	3	1	1	1	4	3		2	1	1	2				2	3		3			1		1	1
Govindraj Nagar	1	1	1	1	1	3			1	1	3	1		1	4	4		1			1	2	2	
Hebbal	2	3		1	2	1			2	1				3		3	1	1	1		1	1		2
Hoskote	1	1	4		1	2		1		3	2	2	2	2	2	2	3	3	2	4			1	1
Jayanagar	1	1		2	1					4	1		1	3	2	2	1			1		2	1	1
Kanakapura	1	1	3	2	2	4		2		2	2		4			1		1	1	1	2	1	3	2
Krishnarajapuram	2	1	4	1		1	3	1	1					1	3	1	1	1	1			1	1	3
Magadi	3		2	1	1	1	3					1	2	2	1	2		1	1	1	1	1	1	1
Mahadevapura	2		2	3	3	2	1	1	2	3	3				1		1	1	1	1	1	1	1	1
Mahalakshmi Layout	1	3	1	1			3	1	2		2	2	1		2	1	1		2	1	1		1	1
Malleswaram			1						1	1	3	1	1		2		1	1	1	1	1	2	1	1
Nelamangala	1		1			1	2	2		4	1	1	2		1	1	1	1	1	1	2	1		2
Other Assemblies	1	1	4		2	2		1			1		3		2	1		2			1	1	1	1
Padmanabhanagar		1	2	3	1		2	2	2		1	2	1		2		2	1		2	3		1	1
Pulakeshinagar		1		3	2	1	2	3		1	1	1	2	1	1	1		2	2			1		
Rajaji Nagar	3	2	2		2	1	1	1	2		1				2	2	1			1	2	1		
Rajarajeshwarinagar	2	3			2	2		3		1			1	1		2	1	1	2			2		5
Ramanagar	3	1	1	2	1		1	1	2	1	1				4			3	1	1	1		1	
Sarvagnanagar	2	1	1	1	2	2		1	2		1	4		1	1				3		2	2	1	2
Shanti Nagar	1		3			4	1	1	2	3	1	1	2	1						2	3	2	1	2
Shivajinagar		3	3		1				3			1					4		1	2			1	1
Vijay Nagar	3	1			1		2	1	2	1	3		2	4	3	1		1	1	2	2	1	1	1
Yelahanka	4	1		1	3	2	2	1	1	2	2	1	1		1			1	1	1	4	2	1	1
Yeshwantpur	2	1	2			2		2		2		1		3		1		1	1					2

(Figure-13)

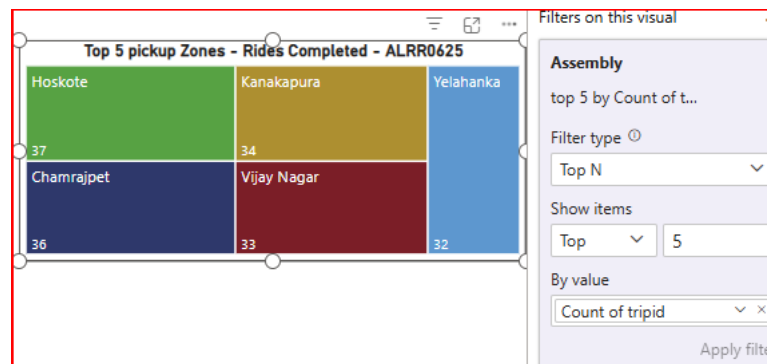
2.8. Top Zones with Highest Trip Volume [3 Marks]

- Identify the top 5 pickup zones with the highest total number of completed trips.
- Analyse factors contributing to the higher number of trips.

Solution:

To identify pickup zones with the highest number of trips completed, we selected assemblies (Pick-up locations) from Assembly table and applied **Top N filter** on number of rides recorded in Trips table.

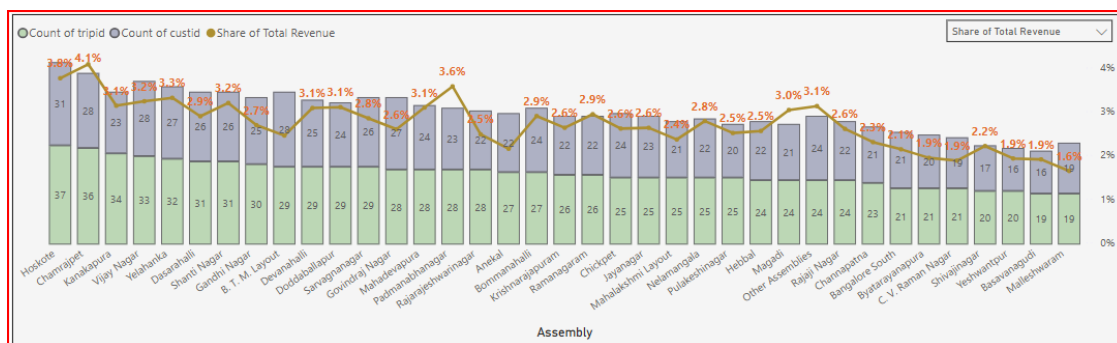
We used Tree plot visual and applied visual level filter to find the following locations with count of rides requested (refer figure -14): -



(Figure-14)

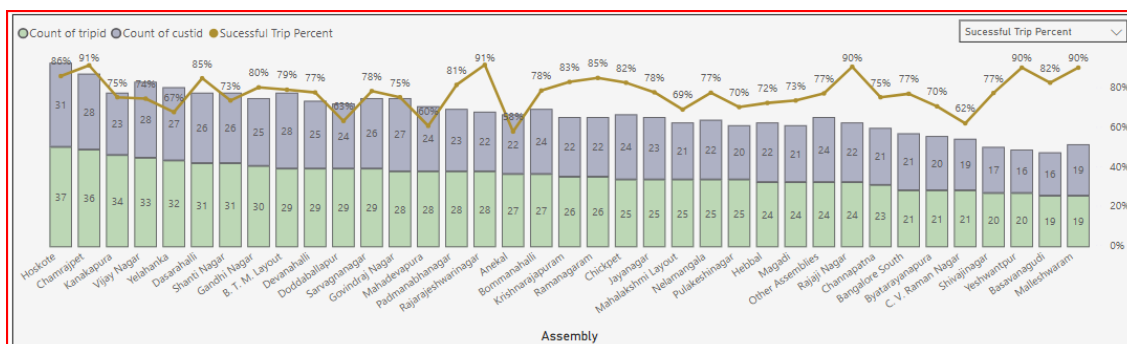
Probable factors contributing to the higher number of trips are: -

- **High Revenue generating zones** – Share of total revenue from these zones is above 3%, with *Hoskote* and *Chamrajpet* weighing above 4%.



(Figure-15)

- **Low Cancellation percentage** – Among top pick-up locations, these locations had high success rate (low cancellation rate)



(Figure-16)

2.9. Basic Analytical Tasks [8 Marks]

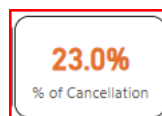
- 2.9.1
What are the percentages of cancellations and successful rides by both driver and customer? [3 marks]

Solution:

To find cancellation percentage by both driver and customer, we used features from Trip_details table and applied following DAX codes to create Total Cancellation Percentage measure: -

```
Trip Cancellation Percent =  
(sum(Trip_Details[searches_got_quotes]) -  
sum(Trip_Details[end_ride]))/sum(Trip_Details[searches_got_quotes])
```

We used a card visual to display the Overall cancellation percentage as a KPI

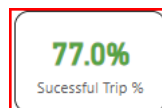


(Figure-17)

Similarly, to find Overall Successful ride percentage by both driver and customer, we used features from Trip_details table and applied following DAX codes to create Total Cancellation Percentage measure: -

```
Successful Trip Percent =  
sum(Trip_Details[end_ride])/sum(Trip_Details[searches_got_quotes])
```

And used it as a KPI again,



(Figure-18)

To find the cancellation and successful ride percentage separately for Customer, we used this DAX code: -

```
Customer Cancellation % = DIVIDE(  
    [Num of cancellation by Customer],  
    sum(Trip_Details[searches_got_quotes])  
)  
  
# Successful Ride  
  
Customer Successful Rides % = 1 - [Customer Cancellation %]
```

Card visual to display Customer cancellation & success percentage as a KPI

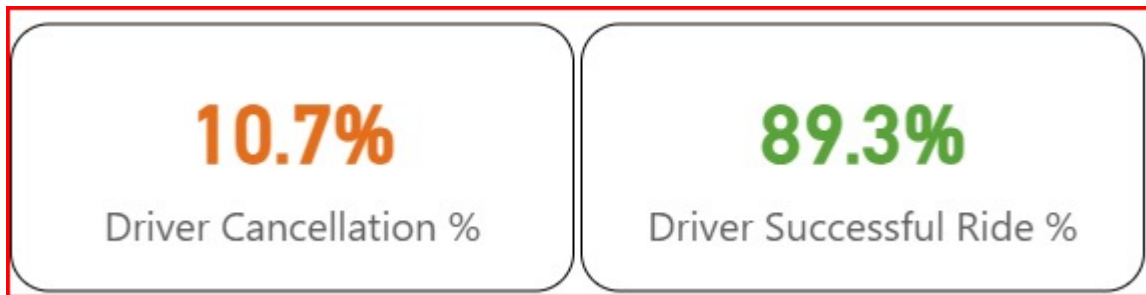


(Figure-19)

To find the cancellation and successful ride percentage separately for Driver, we used this DAX code: -

```
# Cancellation  
  
Driver Cancellation % = DIVIDE(  
    [Num of cancellation by Driver],  
    sum(Trip_Details[searches_got_quotes])  
)  
  
# Successful ride  
  
Driver Successful Ride % = 1-[Driver Cancellation %]
```


Card visual to display Driver cancellation percentage as a KPI



(Figure-20)

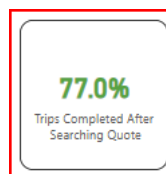
- 2.9.2
Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods.
[5 marks]

Solution:

We derived percentage of people who completed trips after searching for quotes by creating a measure and through a simple funnel chart. The DAX code for the measure used: -

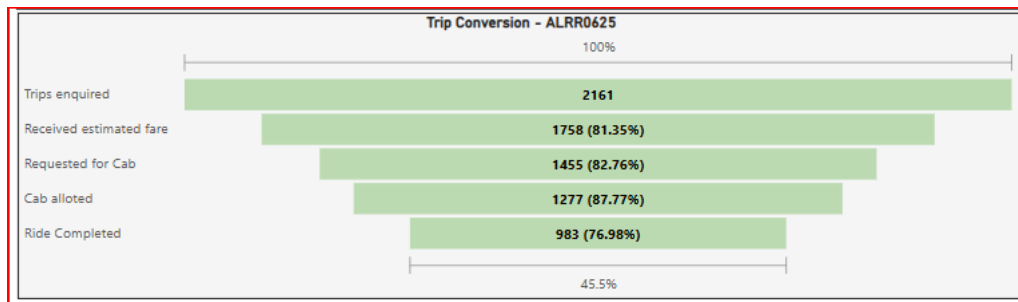
```
Trips Completed After Searching Quote =  
DIVIDE(  
    COUNTROWS('Trips'),  
    SUM(Trip_Details[searches_got_quotes])  
)
```

The percentage is



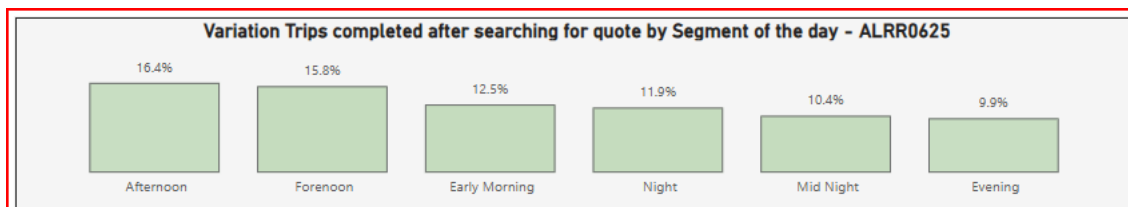
(Figure-21)

The Funnel chart for the same

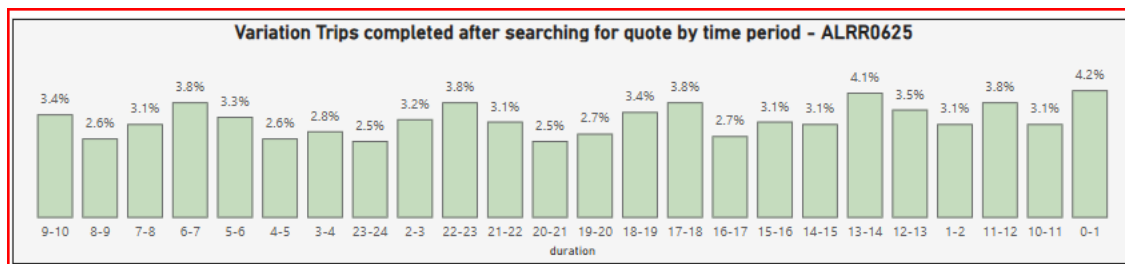


(Figure-22)

To visualize the variation of this ratio by time periods, we simply utilized the above measure on Y-axis and segment of the day/ duration on X-axis on a bar chart. Refer Figure 23.1 & 23.2 for more insights: -



(Figure-23.1)



(Figure-23.2)

2.10. Create a Parameter and Use Filters [5 Marks]

- Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
- Explain your choice of filter and insights drawn from this step.

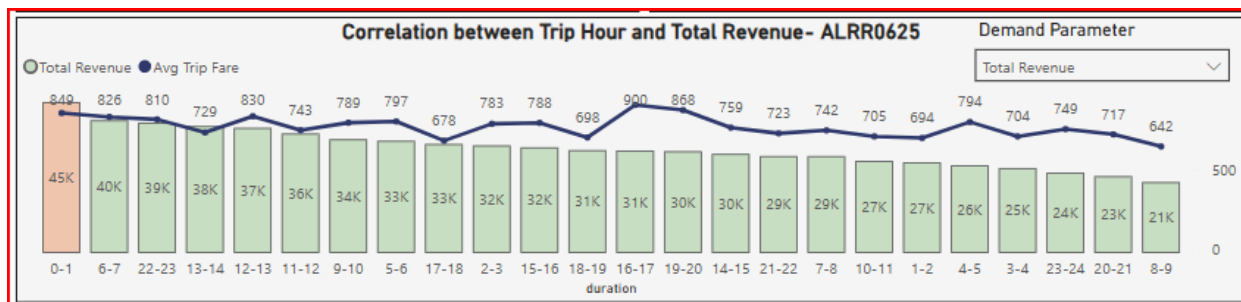
Solution:

We created two Field Parameters, namely – *Demand Parameter* and *KPI Parameter*.

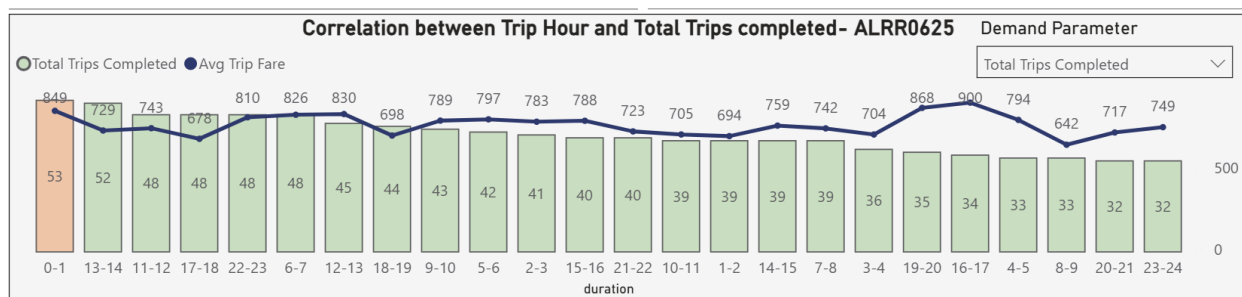
Demand Parameter: To analyze distribution of *Total Revenue* and *Total Trips* count with change in *duration*. This is the autogenerated DAX code for the parameter: -

```
Demand Parameter = {
    ("Total Trips Completed", NAMEOF('Measure_Table'[Total
    Trips Completed]), 0),
    ("Total Revenue", NAMEOF('Measure_Table'[Total Revenue]),
    1)
}
```

Additionally, we can even analyze change in *Average Trip Fare* (placed on secondary Y-axis) with change in duration and other parameter values. Refer figure- 24.1 & 24.2



(Figure- 24.1)



(Figure- 24.2)

We drew following insight: -

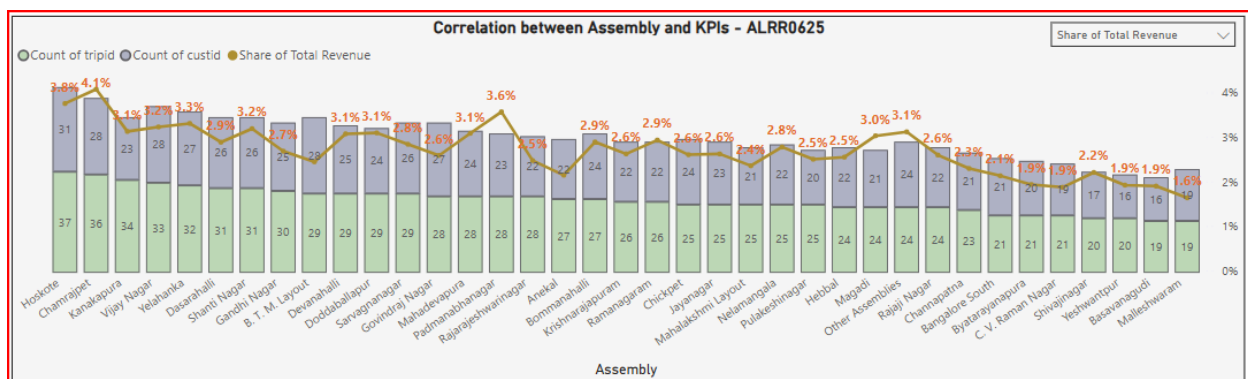
- Hourly correlation of Total Revenue Vs Average Trip Fare: - No significant insights found in this analysis

- Hourly correlation of Total Trip completed Vs Average Trip Fare: - Fare Spikes During Off-Peak Hours - While overall fares are stable, the "Avg Trip Fare" experiences noticeable increases during 16-17, 19-20, and 4-5 hours. These are times when "Total Trips Completed" are relatively lower compared to the peak, suggesting potential surge pricing or higher value trips during these less busy periods.

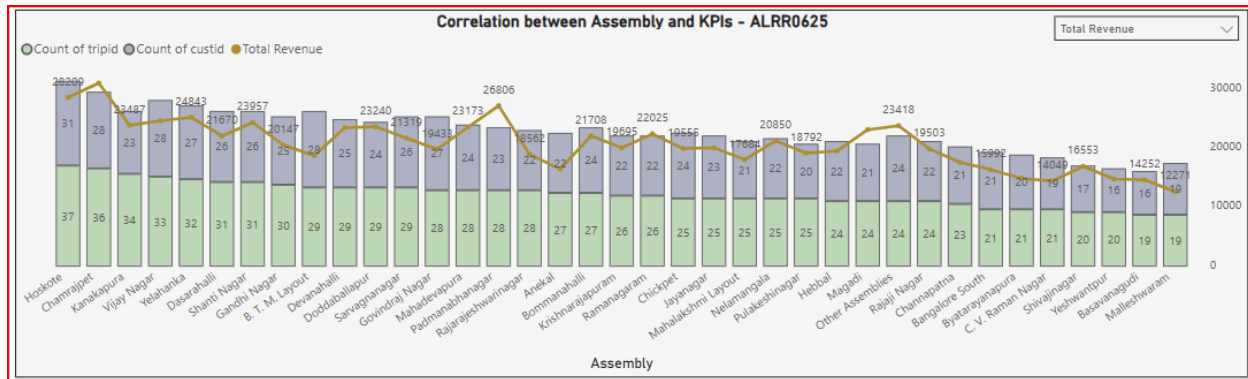
KPI Parameter: To analyze distribution of *Total Revenue*, Share of Total revenue, Trip Rate, Trip Cancellation percentage and Successful Trip percentage w.r.t to each location and number of unique customers. This is the autogenerated DAX code for the parameter: -

```
Parameter = {
    ("Total Revenue", NAMEOF('Measure_Table'[Total
Revenue])), 0),
    ("Share of Total Revenue", NAMEOF('Measure_Table'[Share
of Total Revenue])), 1),
    ("Trip Rate", NAMEOF('Measure_Table'[Trip Rate])), 2),
    ("Trip Cancellation Percent",
NAMEOF('Measure_Table'[Trip Cancellation Percent])), 3),
    ("Successful Trip Percent",
NAMEOF('Measure_Table'[Successful Trip Percent])), 4)
}
```

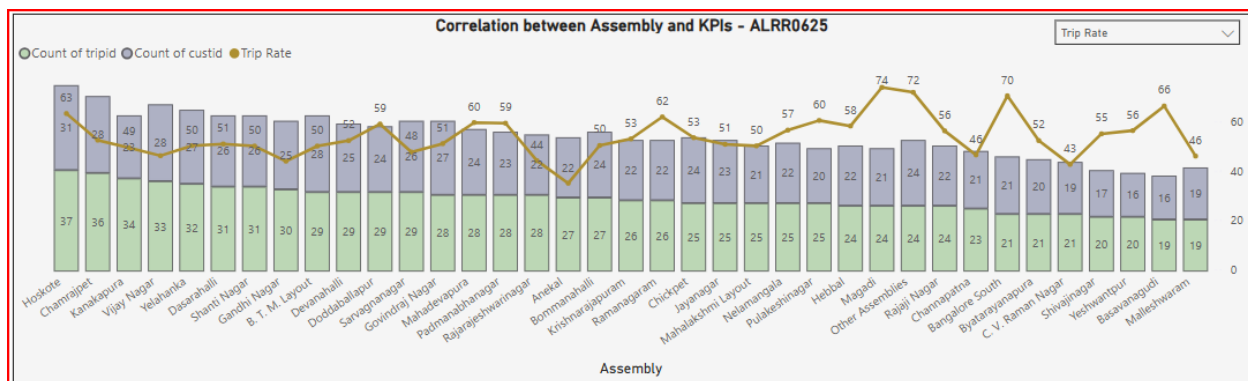
We linked KPI parameter at visual level to filter the *stacked bar chart*.



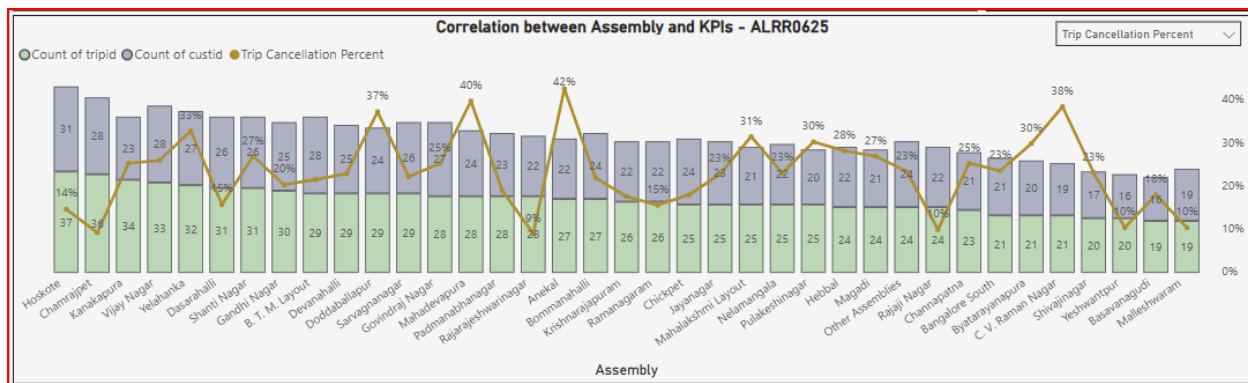
(Figure-25.1)



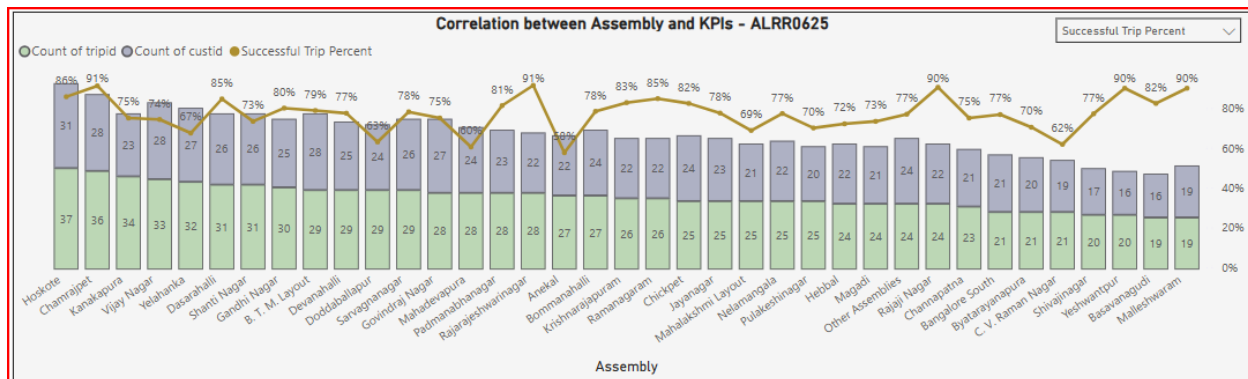
(Figure 25.2)



(Figure 25.3)



(Figure 25.4)



(Figure 25.5)

We drew following insights: -

- Hoskote & Chamarajpet are High-Throughput Regions with Divergent KPIs: Both exhibit high tripid and custid counts. Hoskote shows strong Successful Trip Percent and Total Revenue, while Chamarajpet has a notable Trip Cancellation Percent, indicating efficiency disparities.
- Inverse Relationship Between Success and Cancellation Rates; Revenue Decoupling Observed: Successful Trip Percent inversely correlates with Trip Cancellation Percent. However, Total Revenue does not strictly track the highest Successful Trip Percentages, suggesting other revenue drivers (e.g., average trip value).
- High Trip Rate/Revenue does not guarantee Peak Success Rate: Assemblies can achieve high Trip Rate and Total Revenue without possessing the absolute highest Successful Trip Percent, implying volume or individual trip value contribute significantly.

3. Conclusion

[20 Marks]

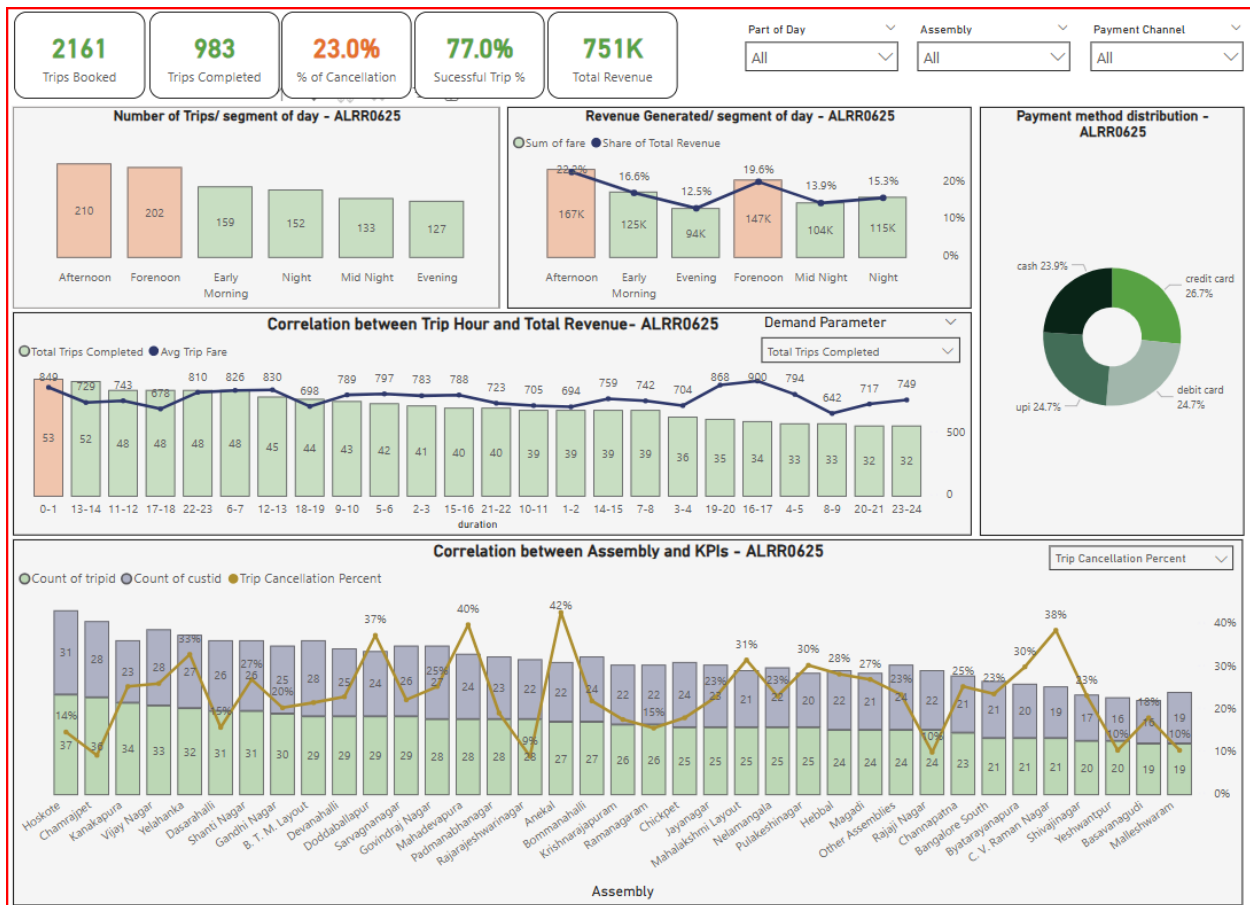
3.1. Recommendations for Operational Efficiency [10 Marks]

- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations.
- Add supporting dashboards.

Solution:

Recommendations for Operational Efficiency are: -

- **Driver Allocation Based on Peak Time Demand -**
 - Peak hours 0–1 AM, 6–7 AM, and 13–14 PM slots show high trip volume.
 - Ensure adequate driver availability during these slots through shift planning or dynamic incentives.
 - Use historical demand (as shown in the dashboard) to schedule drivers proactively.
 - Target top-performing zones with enhanced customer and driver incentives.
 - Monitor quote-to-ride conversion rates to identify friction points in customer experience.
- **Route Optimization for High-Distance Slots –**
 - Average ride distances are higher in early morning and late-night hours.
 - Use this insight to offer flat-fare pricing or bundled discounts to encourage ridership in low-demand periods and improve vehicle utilization.
- **No Cancellations Recorded – But Introduce Monitoring**
 - Improve algorithmic matching during medium-performing time slots to reduce cancellations.
 - Introduce a cancellation early-warning system using factors like driver wait time, customer location mismatch, etc.
- **Reduce Idle Time with Predictive Matching**
 - Use predictive models to assign drivers based on time-slot and zone-specific ride probability.
 - This reduces idle time and improves average trip efficiency.
- **Zone-Level Focus on High Revenue Areas**
 - Prioritize service and promotions in zones that bring in higher average fares (e.g., commercial zones).
 - Consider introducing premium service tiers in those zones.



(Figure-26)

3.2. Marketing and Operational Strategy Improvements [10 Marks]

- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

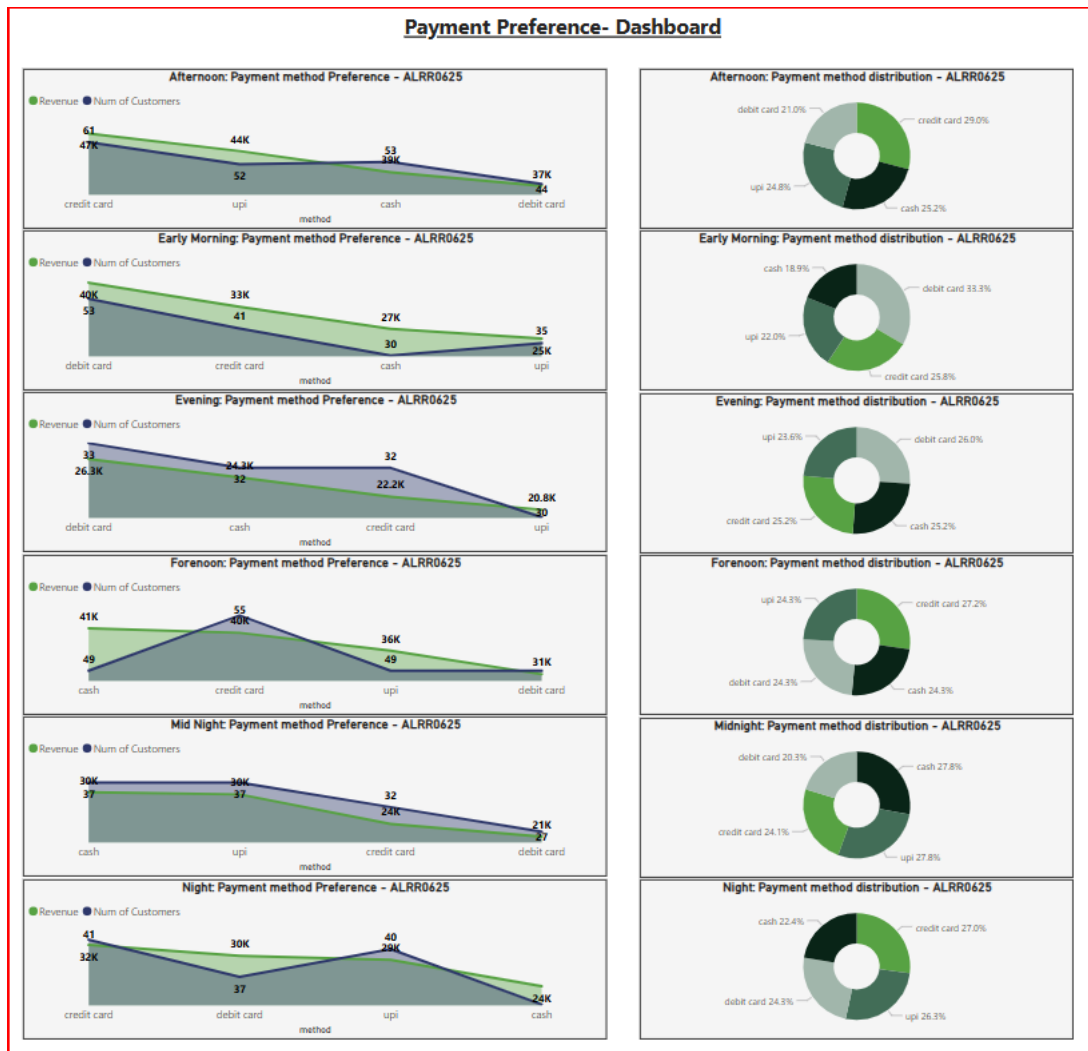
Solution:

Suggestions to enhance Marketing and Operational efficiencies: -

- **Hyperlocal Promotions in High-Traffic Zones**
 - Zones such as Hoskote, Mahadevapura, and Byatarayanapura (as seen in the chart) have both high trip volume and revenue.
 - Launch location-based offers, such as:

- “Ride 3 times this week from Mahadevapura, get ₹50 off”
- Geo-fenced push notifications
- Introduce loyalty programs in zones with high trip completions.
- **Time-Specific Discounts**- Offer incentives during low-demand hours (1-6 AM and 1-4 PM): Example: “Flat ₹30 off on rides between 2-4 PM”. This encourages off-peak usage and boosts utilization rates.
- **Promote Preferred Payment Methods** - UPI and Credit Cards are dominant — run campaigns like: “Get ₹20 cashback on your first 3 UPI rides”. Incentivizes digital payment adoption and improves transaction reliability.
- **Corporate & Institutional Tie-ups** -Zones near tech parks and business hubs can be targeted for: Monthly ride pass subscriptions. Priority pickup for office commutes
- **Loyalty Program for Frequent Riders**- Build a customer loyalty program offering: Free rides after 10 trips. Tiered benefits (Silver, Gold, Platinum)
- **Real-Time Feedback Collection**- Encourage users to share feedback post-ride — especially in high-volume zones — for continuous service refinement.

These strategies, aligned with operational patterns, can help Namma Yatri boost user engagement, revenue, and service quality across key segments.



(Figure-27)